

## Task 1: Comparison of Operating Systems

| Operating System | Main Features                                                                               | Target Devices                               | Examples of Apps                              |
|------------------|---------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------|
| Windows          | User-friendly interface, multitasking, wide hardware/software compatibility, gaming support | Desktop computers and laptops                | Chrome, MS Office, VS Code, Instagram, Zomato |
| Linux            | Open-source, highly customizable, secure, powerful command-line tools                       | Servers, desktops, laptops, embedded systems | Firefox, VLC, Terminal, VS Code               |
| Android          | Touch-based interface, mobile app ecosystem, notifications, Google services                 | Smartphones, tablets, smart TVs              | WhatsApp, Instagram, Paytm, Zomato            |
| IOS              | Strong security, controlled app ecosystem, smooth hardware/software integration             | iPhone and iPad                              | WhatsApp, Instagram, YouTube, Paytm           |

## Task 2: Operating Systems on My Devices

Phone: Android

- My phone runs Android.

Two features that affect my daily use of apps:

1. App permissions: Android allows me to control permissions such as camera, microphone, location, contacts, and storage for individual apps. This helps control what apps can access.

2. Background app management: Android manages applications running in the background to reduce battery and memory usage. This can affect how apps such as Instagram, WhatsApp, or Zomato receive notifications and update their content.

### Task 3: Security Risks in Mobile Operating Systems

| Security Risk             | Impact on Apps                                                                                                                                                                    |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Malicious or fake apps    | A fake version of an app such as WhatsApp or Paytm could contain malware and attempt to steal personal information or login credentials.                                          |
| Excessive app permissions | A malicious app with unnecessary access to SMS, contacts, microphone, or files could collect sensitive information and potentially expose data used by other applications.        |
| Outdated operating system | If security updates are not installed, attackers may exploit known vulnerabilities. This could compromise personal data or information handled by apps such as WhatsApp or Paytm. |

### Task 4: Cloud-Based Operating System vs Traditional Desktop OS

A traditional desktop operating system, such as Windows, normally runs most of the operating system and applications directly on the computer. Applications and files can be stored locally on the device.

A cloud-based operating system, such as ChromeOS, is designed around web applications and cloud services. Many applications and files can be accessed through the Internet, although ChromeOS also supports local applications and offline functionality.

| Traditional Desktop OS                                                 | Cloud-Based OS                                                          |
|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Applications can be installed and run locally.                         | Many applications are accessed through web/cloud services.              |
| More software and files can be stored directly on the computer.        | More reliance on cloud storage and Internet services.                   |
| Can often operate extensively without an Internet connection.          | Many cloud-based services work best with an Internet connection.        |
| Usually requires more local system resources for desktop applications. | Can provide a lightweight experience focused on web/cloud applications. |

Example: YouTube or Flipkart

On a traditional Windows laptop, I can open Chrome and access YouTube or Flipkart through the browser, while other desktop applications can run locally at the same time.

On a cloud-focused device such as a Chromebook, I would typically access YouTube or Flipkart through the browser as web services. My account, preferences, and cloud-stored information can be accessed from another compatible device after signing in.